ASSIGNMENT - PHL509/PH509

Based on lecture note 2

Group EE1 (page 2-4)
Explain the four fundamental forces in nature and their interaction particles.
EE2 (page 1-3)
Particle classification in the standard model of particle physics.
ME1 (page 5)
Describe the basic nuclear properties. Explain nuclear magnetic moment.

Based on lecture note 3

ME2 (page 6)
Derive and describe the hydrodynamical equilibrium used to explain the stability of white dwarfs
CS1 (page 2-3)
The nucleus can be modelled as a degenerate gas of fermions. Write the thermal distribution and obtain the grand potential and the number density.
CS2 (page 5)
The nucleus can be modelled as a degenerate gas of fermions. Obtain the expressions for number density, energy density and average energy considering a non-relativistic energy dispersion relation E=p^2/2m.
DSAI+MSME (page 5)
The nucleus can be modelled as a degenerate gas of fermions. Obtain the expressions for number density, energy density and average energy considering a relativistic energy dispersion relation E=sqrt(p^2c^2+m^2c^4).
MT (page 5)
The nucleus can be modelled as a degenerate gas of fermions. Obtain the expressions for number density, energy density and average energy considering a relativistic energy dispersion relation E=sqrt(p^2c^2+m^2c^4) in the limit m tends to zero.